

UNIVERSITY OF SOUTHAMPTON
Faculty of Engineering and Physical Sciences
School of Electronics and Computer Science

A project report submitted for the award of
MEng Biomedical Engineering (AI)

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**AI-Enhanced Multi-Functional
Metalens Microscope for Cellular
Abnormality Identification**

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ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
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Fourier light-field microscopy (FLFM) supports the development of more compact and computationally efficient imaging systems compared to current approaches. However, accurate reconstruction of a 3D sample volume from a 2D measurement remains a difficult inverse imaging problem that needs to be solved. This project focused on developing and comparing four physics-based FLFM reconstruction methods: a baseline Richardson-Lucy (RL) reconstruction algorithm, an RL method with a total variation (TV) regularisation term, a simplified 3D Gaussian reconstruction algorithm inspired by 3D Gaussian adaptive tomography (3DGAT), and a TV-regularised simplified 3D Gaussian method. Simulated point spread functions (PSFs) and bead test data were used within a shared FLFM framework implemented in Python. The reconstruction quality of the proposed algorithms was evaluated using Sparrow separability, full width at half maximum (FWHM), and algorithm runtime analysis. In addition to this, a basic XAI application was developed to visualise reconstruction outputs and improve interpretability. The overall results showed that the RL + TV algorithm provided the strongest overall balance between reconstruction quality, localisation performance, and computational costs. The simplified Gaussian method exhibited some competitive Sparrow separability performance, but at a substantially higher computational cost without clear FWHM improvements. The simplified Gaussian + TV method performed the worst overall, generating reconstructions dominated by arbitrary noise. The findings from this project show that for evaluating FLFM reconstruction methods, multiple qualitative and quantitative metrics are required to understand the trade-offs between localisation performance, separability performance, and computational efficiency. Overall, these findings suggest that the RL + TV algorithm was the most favourable out of all the evaluated methods.

Statement Of Originality

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I confirm that I have not previously submitted any part of this work for another assessment, except where explicitly permitted for this module. For example, re-using materials from the progress report, etc. My work did not involve human participants, their biological material or personal data, or animals.

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List of Symbols

d_s	Sparrow resolution limit (minimum resolvable separation)
λ	Wavelength of the light used in imaging
NA	Numerical aperture of the objective lens
O	Measured 2D FLFM image
$\hat{O}$	Predicted FLFM measurement
g	Reconstructed 3D object volume
$g^{(k)}$	Reconstruction estimate at iteration k
H	Forward imaging operator / PSF operator
H_j	PSF slice at depth j
k	Iteration number
FWHM	Full width at half maximum
x_{left}	Left position at half maximum
x_{right}	Right position at half maximum
M	Number of Gaussian components
ρ_i	Learnable intensity weight of the i -th Gaussian
$\mu_{z,i}, \mu_{y,i}, \mu_{x,i}$	Centre coordinates of the i -th Gaussian
$\sigma_{z,i}, \sigma_{y,i}, \sigma_{x,i}$	Widths of the i -th Gaussian along the z , y , and x axes
$g_\theta(z, y, x)$	Gaussian-parameterised reconstruction volume
$g_{\theta,j}$	Reconstructed object slice at depth j
λ_{TV}	TV regularisation weight
L_{Total}	Total loss function
L_{ImgDData}	Image-domain data term
L_{TV}	Total variation penalty term
$P(g)$	TV regularisation term applied to the reconstruction

Chapter 1: Introduction

Currently, there are only limited specialist microscopes that can help identify and understand cellular abnormalities within the medical and biomedical research sectors. These dated microscopes typically take up large amounts of space and are very costly. As a result of this, access to these technologies is greatly reduced for many research laboratories and hospital environments. As technology develops, more robust and compact methods are needed to identify cellular abnormalities that integrate current technologies and computational techniques, such as the use of artificial intelligence.

This project aims to make significant contributions to the development of a metalens-based microscope, specifically focusing on the development of the software creating a pipeline needed to make this microscope work with strong accuracy, reliability, consistency, reusability, and repeatability.

Chapter 2: Literature Review

2.1 Fourier Light Field Microscopy

Fourier light-field microscopy (FLFM) is an extension of light-field microscopy (LFM) where FLFM maps recorded light information to the Fourier domain instead of directly forming images through the native image plane [1]. The Fourier domain is where information is processed into spatial information and then expressed as spatial frequencies. This enables a better description of light propagation for image formation. Due to the different approaches from LFM, FLFM allows high-resolution imaging across two or three-fold extended depths by mitigating artefacts [1]. Guo et al. implemented FLFM on a conventional epifluorescence microscope using a $40\times/0.95 \sim \text{NA}$ objective lens, where a Fourier lens was used to perform an optical Fourier transform of the images on the native image plane. At the back of the Fourier lens, a microlens array was placed, and the light field was imaged using a 1:1 relay lens, recorded on a scientific complementary metal-oxide-semiconductor (sCMOS) camera [1]. The following image explains the experimental setup described:

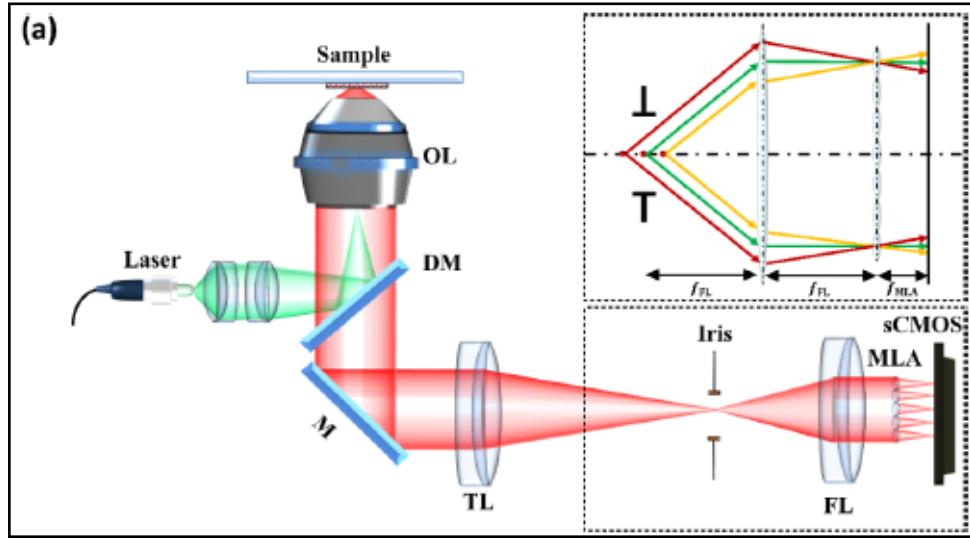


FIGURE 2.1: Optical layout of the Fourier light-field microscopy (FLFM) system, showing the excitation path, objective lens (OL), tube lens (TL), Fourier lens (FL), microlens array (MLA) and sCMOS camera. Taken from Guo *et al.* [1].

With this setup, results such as faster computational times, higher resolutions and better noise and artefact reductions were obtained. These results make FLFM important for this project, not just for its powerful imaging potential for cellular imaging, but also

for the ability to extend the depths with higher resolutions and shorter computational times [1]. This therefore reduces computational costs, noise and artefacts, and outperforms traditional light-field microscopy approaches.

2.1.1 Reconstruction Algorithms

In FLFM, image reconstruction is treated as a deconvolution problem. This is because the recorded light-field image is described by a forward model that convolves the 3-dimensional (3D) original scene (what the microscope is viewing as an unknown 3D sample volume) with an associated point spread function (PSF).

In this project, four image reconstruction strategies are explored: a baseline Richardson-Lucy image reconstruction algorithm (RLA) used in the original FLFM by Guo et al. [1], an RLA with a regularisation term of total variation (TV), 3D Gaussian image reconstruction, and 3D Gaussian reconstruction with a TV regularisation term. To evaluate the effectiveness of different image reconstruction algorithms, metrics from tests such as Sparrow separability and full width at half maximum (FWHM) are used, along with other performance metrics such as runtime and hardware usage.

2.1.1.1 Richardson-Lucy Image Reconstruction Algorithms

Guo et al. [1] use the Richardson-Lucy image reconstruction algorithm to solve this deconvolution problem. The RLA begins with an initial estimate of the original scene and then iteratively updates this estimate, making the predicted image (from the FLFM forward model) more consistent with 2D light-field images captured from the sCMOS camera [1]. The goal of this is to recover a 3D reconstruction that explains the measured 2D data with minimal noise and artefacts.

However, for the RLA, there are some limitations to consider. One limitation is that if there are many iterations, then this has a chance to significantly blow up the noise, resulting in less effective image reconstructions [2]. Other limitations include high sensitivity to PSF inaccuracies and poor adaptability to blind deconvolution scenarios [2]. Some ways in which these limitations can be addressed are to use a small number of iterations and implement regularisation terms.

2.1.1.2 3D Gaussian Image Reconstruction Algorithms

The RLA algorithm has poor axial resolution and significant signal loss during reconstruction [1], which makes it difficult for accurate and precise 3D reconstruction at high resolution. According to Xu et al., 3D Gaussian Adaptive Tomography (3DGAT) improves reconstruction accuracy and resolution relative to conventional deconvolution-based methods [3].

In 3DGAT, the reconstruction target is represented as a set of 3D Gaussian kernels, where each kernel is a 3D fluorescent intensity field in the 3D space. The 3D Gaussian kernels have learnable parameters and represent arbitrary ellipsoids [3]. The 3DGAT method first performs a Wiener reconstruction to obtain a coarse volume, which is then used to initialise 3D Gaussian kernels. This is done by applying a Wiener filter to the known 2D FLFM measurement, which is the input data to the reconstruction [3].

An intensity voxeliser then converts the Gaussian representations into the voxel representations. This dense volume of the voxel representations is then used for projection in the forward model. The Gaussian parameters are then optimised through gradient descent with a goal of minimising reconstruction loss. In the original 3DGAT method, the loss function is a combination of image domain Mean-Squared-Error (MSE) and frequency domain loss [3]. In this project, a simplified 3D Gaussian reconstruction algorithm, which is inspired by 3DGAT, will be considered. This algorithm will then be extended by adding a regularisation term and compared against the other proposed reconstruction algorithms.

The main difference between the RLA algorithm and 3DGAT is therefore the way in which the algorithms represent objects and their optimisation strategies. RLA reconstructs a dense voxel volume directly. In contrast, 3DGAT reconstructs a set of Gaussian components that are then voxelised and projected through the same FLFM imaging model [1, 3].

In general, Gaussian based scene representations gained popularity through 3D Gaussian Splatting and were subsequently adapted to solve inverse imaging and tomographic reconstruction problems [4, 5]. For the inverse imaging context, 3DGAT applies a Gaussian representation specifically to FLFM reconstruction under the same forward model [3]. This allows for 3D Gaussian imaging reconstruction approaches to be suitable and relevant alternatives to voxel-based Richardson-Lucy algorithms, specifically when trying to understand and compare the reconstruction quality, computational performance under a the same FLFM forward model, and axial localisation (how accurate the localisation of structures along the z-axis is from reconstruction) [4, 5].

2.1.2 Regularisation Terms

For FLFM reconstructions, regularisation terms are crucial as they help combat the limitations faced by the baseline RLA. Many types of regularisation terms help reduce the algorithm’s sensitivity to noise. One regularisation term that does this is Total Variation (TV) regularisation. TV penalises the spatial gradient of the image reconstruction by encouraging piecewise-constant structures and, as a result, suppressing noise and preserving sharp edges within the image [6, 7]. When TV is combined with the RLA, the regularisation term helps to act as a denoiser. This TV prior reduces the risk of overfitting from noise and minimises oscillatory artefacts [7]. However, trade-offs can come from using the TV regularisation term, one of which is that there is a risk of over-smoothing precise information if the weight of the TV prior is too strong [7]. Evan [8] provides a useful baseline for incorporating a TV regularisation term into the RLA. This paper also provides motivation for the TV prior to be used as a regularisation term for reducing noise amplification. In this project, the idea is extended to the simplified 3DGAT approach by adding a TV penalty in the optimisation loss process.

Other regularisation approaches for inverse imaging problems, such as convolutional neural network plug-and-play (CPnP) regularisation, have been explored. CPnP is a powerful denoiser where the gradient step associated with a prior is replaced by an image denoising algorithm [9]. This image denoising algorithm defines a learned regulariser. When combined with the RLA, high-frequency features are preserved, and noise and artefacts are removed [9]. However, this project will focus on the TV regularisation term and its effects on different reconstruction algorithms.

2.2 Explainable AI (XAI)

Explainable AI refers to the methods used to make Artificial Intelligence interpretable and more understandable to humans [10]. In contrast to black-box models, where no one fully understands the thought process, XAI techniques aim to describe these unknown thought processes and also explain them in visual ways, highlighting things like what regions or input patterns impact and determine the outputs. For this project, I will be focusing on visualisation methods to explain the algorithms. However, there are many XAI techniques, such as simulations of neural networks, attribution maps, and surrogate models that approximate complex networks with simpler interpretations [11, 12].

For medical imaging, it is important to use XAI so that clinicians and researchers can understand why an algorithm focuses on certain structures or features before its results

can be confidently used to investigate patient problems [13]. Another consideration is that currently, there is a lot of reluctance and a lack of adoption for using AI and technologies that people do not understand, so by applying XAI techniques, this helps to bridge the gap and improve adoption rates for these technologies.

2.3 Optical Simulation

Optical Simulations are used to represent how a microscope would form an image [14]. Through optical simulation, PSFs are computed numerically when they cannot be measured experimentally [15]. For FLFM, accurate PSFs are crucial for defining the forward model, as without them, the deconvolution-based reconstruction would not be possible [1]. This means that PSFs which cannot be obtained experimentally, simulated ones act as alternatives. For this project, optical simulation will be used to generate the PSFs required for FLFM, which are required for the reconstruction algorithms to create an artificial bead dataset for Sparrow separability and FWHM testing which also contributes towards the basic XAI application.

2.4 Summary and Identified Gaps

This literature review has outlined the key concepts needed to ensure that planning, implementation and testing stages for this project can be carried out effectively. The literature on optical simulation highlights why generating PSFs artificially is important and shows how they can be used to form part of an XAI basic application, where they are needed for reconstruction demonstrations. The literature shows that it is important to implement XAI to increase the adoption of technologies that use artificial intelligence and machine learning systems. Researching and contrasting FLFM and Light Field Microscopy (LFM) has highlighted that there is limited accessibility to high-complexity microscopes. These have the potential to be redesigned to reduce costs, space, computational times, and increase accessibility in hospitals and research labs. From this research, it appears that there are currently limited systematic comparisons of FLFM reconstruction methods. Current research mainly focuses on developing or improving individual algorithms instead of directly comparing different algorithms using consistent resolution, runtimes, and reconstruction quality metrics. These identified gaps motivate this project to evaluate the current state-of-the-art algorithms using a shared FLFM forward model and common performance measures.

Chapter 3: Requirements and Specification

3.1 Project goals and Problem Definition

The main problem that is addressed in this project is solving the inverse imaging problem. Reconstructing a 3D sample volume from a 2D FLFM measurement and a known PSF effectively is challenging. This is because different reconstruction methods have different trade-offs in resolution, axial localisation, and computational costs.

The overall goal of the project is to reconstruct a 3D sample volume from a 2D FLFM measurement using physics backed reconstruction approaches and then visualise the reconstruction outputs through an XAI-based application. The purpose of using an XAI application is to qualitatively visualise outputs with a goal of increasing adoption and interpretability of the applications if implemented in clinical settings. The project framework has been designed to implement an RLA and an extended RLA with TV regularisation, and to evaluate these algorithms using Sparrow separability testing, FWHM and performance runtimes. The project framework has also been designed to implement a simplified 3D Gaussian reconstruction method inspired by 3DGAT and an extended version of this incorporating TV regularisation. These four image reconstruction approaches are specified for comparison in order to understand and investigate the effects of object representation, TV regularisation, computational performance, and optimisation techniques on reconstruction quality.

3.2 Functional Requirements

The functional requirements have been listed below to clearly highlight the core functionality that is expected from work carried out within this project. Requirements were prioritised as either must or should requirements. Must requirements define the essential functionality, and should requirements describe the desirable functionality that supports the overall project goals.

3.2.1 Reconstruction Algorithm Requirements

1. The framework needs to implement the baseline RLA for FLFM reconstruction.

2. The framework needs to implement RLA with a TV regularisation term.
3. The framework needs to implement a simplified 3D Gaussian reconstruction method inspired by 3DGAT.
4. The framework needs to implement a TV-regularised version of the simplified 3D Gaussian reconstruction method.
5. The framework should allow reconstruction parameters, such as iteration count and regularisation weight, to be adjusted. This is to find representative benchmark settings for different reconstruction approaches by finding near-best performances from the different baseline algorithms.

3.2.2 Input and Output Requirements

1. The framework needs to accept simulated PSFs and the relevant corresponding FLFM measurement images or test objects as inputs.
2. The framework needs to output 3D reconstructed volumes or 2D reconstructed images dependent on the PSF and FLFM input measurements.
3. The framework needs to handle grayscale images as a minimum as an input object.
4. The framework needs to output grayscale images as a minimum.
5. The framework should allow for saving reconstructed volumes for later evaluation and visualisation.
6. The framework needs to support input data stored in .mat format.
7. The framework should support multiple output formats that are suitable for volumetric analysis. For example, TIFF stacks.

3.2.3 Testing and Metric Evaluation Requirements

8. The framework needs to compute the Sparrow separability for pairs of beads at different distances of separation in the reconstructed images.
9. The framework needs to compute FWHM for reconstructed bead profiles to evaluate the axial localisation performance.

10. The framework needs to compare Sparrow separability across the four different reconstruction approaches: RLA, RLA with TV regularisation, simplified 3D Gaussian approach, and simplified 3D Gaussian approach with TV regularisation.
11. The framework needs to compare FWHM results obtained from the four proposed reconstruction algorithms.
12. The framework should evaluate the computational costs and runtime efficiency per algorithm, for different image resolutions.

3.2.4 Explainable AI (XAI) Requirements

13. The basic application framework needs to take a pre-generated PSF and a corresponding input image as a minimum.
14. The basic application framework needs to demonstrate all four reconstruction approaches as a minimum.
15. The basic application framework should highlight whether the reconstruction was resolvable or not.
16. The basic application framework needs to provide an interface that allows users to select the reconstruction algorithm as a minimum.
17. The basic application framework needs to visually compare the original against the reconstructed image as a minimum.
18. The basic application should show intensity profiles where possible.

3.3 Non-Functional Requirements

For this project, multiple non-functional requirements must be considered in addition to the functional requirements previously described. These non-functional requirements relate to reliability, maintainability, usability, reproducibility, and computational practicality. The proposed requirements are extremely important for ensuring that the project provides a consistent framework. A framework in which the behaviour of different reconstruction approaches can be compared fairly under the same FLFM forward model.

The non-functional requirements are as follows:

1. Reliability: The framework must appropriately handle valid inputs consistently without unexpected failure during normal use.
2. Reproducibility: Using the same inputs and parameter settings should produce the same reconstruction outputs and evaluation results. For example, the same PSF stack and the corresponding FLFM measurements as inputs for a specific reconstruction algorithm should produce the same output each time.
3. Maintainability: The code should be built and organised in a modular way such that changing the reconstruction algorithm, testing approach and visualisations can be done independently.
4. Usability: The interface of the XAI basic application should be simple and clear to use so that reconstruction outputs and visualisations of the algorithms can be done easily. Additionally, Python scripts that were used for reconstruction, testing, and evaluation should have steps that are well structured so that they can be clearly understood without misinterpretation and reused easily.
5. Computational practicality: The memory usage and the runtime of the project framework should be manageable on the available hardware.

3.4 Hardware and Software Constraints

The project is subject to a number of hardware and software constraints that influence the scale and capability of the reconstruction experiments that can be carried out. Specifically, the analysis of high-resolution volumetric data is limited by the available computing resources. Alongside this, careful consideration was required when selecting appropriate implementation tools to ensure that they remained usable under these constraints.

The code and application development for this project was carried out using a home desktop PC equipped with an Intel Core i7-13650HX CPU, 24 GB RAM, and an NVIDIA RTX 4060 GPU with 8 GB of dedicated memory. This hardware was sufficient for implementing and testing the different reconstruction algorithms. However, RAM and GPU limitations restricted the practicality of processing very high-resolution images and large PSF stacks in a timely manner. These constraints were problematic for repeated experiments and computationally intensive reconstruction methods. The project framework was implemented mainly using Python. However, Python is less computationally efficient than other low-level languages such as C or C++, which increases the runtime for iterative reconstruction and large-scale volumetric processing.

3.5 System Evaluation

The project framework must be evaluated in a way that ensures the proposed reconstruction approaches can be compared under the same FLFM forward model fairly and consistently. Therefore, the evaluation is based on two main aspects: computational costs for practical use cases and reconstruction quality in terms of resolution and relevant localisation performance metrics. Evaluating these aspects ensures that the work done for this project can be compared and aligned with expected goals.

The computational costs of the framework should be evaluated by considering the runtime performance during reconstruction. This is important because FLFM reconstruction is iterative and involves repeated convolution or projection operations across depth dependent PSF stacks, which can become rapidly more computationally expensive for larger volumes [3]. Therefore, to determine whether the reconstruction methods are suitable for repeated experimentation and comparative testing on available hardware, the computational costs should be assessed.

The final aspect that must be evaluated for this project is the quality of the reconstructions. These evaluations can be carried out using two different testing approaches, Sparrow separability and full-width at half maximum (FWHM) testing. Sparrow separability testing is used to determine the minimum separation at which two reconstructed beads can remain distinguishable. Sparrow separability is particularly useful for this project because it provides a direct and interpretable measure of the lateral resolving capability from bead-based simulations. This means that fair comparisons can be made between different reconstruction approaches as they are all tested under the same controlled conditions. FWHM testing is used to gauge the depth localisation performance from reconstruction. Both of these testing approaches can provide a quantitative basis for comparing the reconstruction accuracy and algorithm behaviour. Overall, these evaluation measures help to provide a basis for comparing reconstruction behaviour alongside localisation performance and computational costs across the four reconstruction algorithms.

Chapter 4: System Design

All four reconstruction algorithms were designed using a shared FLFM pipeline such that fair comparisons could be made under the same input conditions and forward imaging model. Across all methods, the measured FLFM image and corresponding PSF were loaded from `.mat` files and converted to `float32` to standardise inputs. After this, dimensionality checks were performed, and the inputs were then processed using the same Nyquist-style low-pass filtering, clipping to remove negative values, and PSF normalisation steps. The same FLFM forward model, which is depth dependent, was then used for reconstruction. The main differences between the four reconstruction methods come from how the unknown 3D object volume is represented and how the reconstruction update and optimisation are performed.

4.1 Design of Base Reconstruction Algorithm

The baseline Richardson-Lucy reconstruction algorithm was designed as a voxel-based iterative deconvolution method. This approach enables recovery of 3D sample volumes from 2D FLFM measurements using a known depth dependent PSF stack. Using this design, the 2D measured FLFM image is the observed output of the forward imaging model. Additionally, the unknown 3D object volume is the quantity to be reconstructed. The forward imaging model is expressed as the sum of convolutions between each depth slice of the object volume and its corresponding PSF slice.

For this method, the main baseline design is the use of a voxel-based iterative Richardson-Lucy update. This approach kept the same shared preprocessing pipeline and depth dependent forward imaging model described in the prior part of this chapter.

After initial preprocessing has been done, an initial reconstruction estimate is generated by convolving the measured 2D FLFM image with each flipped PSF slice. The PSF stack is spatially flipped to form back-projection kernels. Over 20 iterations, the reconstructed volume is then iteratively updated. After each iteration has been completed, a predicted 2D image is formed by summing the convolutions of all the reconstructed depth slices with their corresponding PSF slices. A ratio image is then calculated between the predicted and measured images. A small stabilising constant was added to this calculation to avoid divisions by zero. The calculated ratio image is then back-projected through the flipped PSF slices and applied as a multiplicative update alongside a pre-calculated normalisation

term. After each iteration, non-negativity was also enforced to prevent instability. After all 20 iterations are completed, each reconstructed slice is filtered again, and the final volume is spatially flipped then saved as a TIFF stack for later evaluation.

Overall, the baseline reconstruction follows the Richardson-Lucy update logic used in FLFM reconstruction. This can be written as:

$$g^{(k+1)} = \text{diag} \left[\text{diag} (H^T H g^{(k)})^{-1} (H^T O) \right] g^{(k)} \quad (4.1)$$

where $g^{(k)}$ is the current reconstruction estimate, O is the measured FLFM image, H is the forward imaging operator, and k is the iteration number [1].

In this project, this logic was implemented through an equivalent slice-wise convolution with ratio image computations and back projections across the depth dependent PSF stack.

4.2 Design of Base Algorithm with TV Regularisation Term

This algorithm is an extended version of the baseline Richardson-Lucy reconstruction algorithm proposed in Section 4.1, where a regularisation term, known as Total Variation, was incorporated. This was added to the iterative update, and the purpose of this extension was to reduce noise amplification and improve the overall stability of the reconstruction whilst keeping the same shared FLFM forward model used in the baseline design. This allows for fair comparisons between the baseline algorithm and the regularised, version where the effects of a regularisation term can be explored whilst preserving the underlying pipeline.

This approach kept the same preprocessing pipeline and FLFM forward model described earlier. This was done so that the effect of the TV regularisation could be evaluated fairly against other methods.

The main extension of this algorithm is the introduction of a curvature-based TV term calculated from the current reconstruction estimate. Following the anisotropic TV regularised Richardson-Lucy formulation, the baseline multiplicative update is kept, but an additional TV term is introduced in the denominator. For clarity, the following equation

is adapted from Wetzstein [16] and rewritten using notation consistent with Section 4.1:

$$g^{(k+1)} = g^{(k)} \cdot \frac{H^T \left(\frac{O}{Hg^{(k)}} \right)}{H^T \mathbf{1} + \lambda P(g^{(k)})} \quad (4.2)$$

where the TV regularisation term is written as

$$P(g) = D_x^T \frac{D_x g}{|D_x g|} + D_y^T \frac{D_y g}{|D_y g|} \quad (4.3)$$

In this project, an equivalent slice-wise form of this update was implemented using a curvature-based TV term, spatial gradients within the plane, and a regularisation weight applied inside the multiplicative Richardson-Lucy update. Since the gradients are computed only in-slice spatial directions, instead of implementing a fully isotropic 3D TV regularisation term, a slice-wise spatial TV term is implemented. This approach is consistent with the role of the TV prior in the cited formulation, where the added denominator term penalises variation during iterative reconstruction, which is unstable.

4.3 Design of Simplified 3D Gaussian Adaptive Reconstruction inspired by 3DGAT

The simplified 3D Gaussian reconstruction algorithm was inspired by 3DGAT and, for this project, was simplified specifically by not including rotational terms or a full covariance matrix in the axis-aligned Gaussian representation. The main decision for this algorithm design was to act as an alternative to the currently seen voxel-based approaches described in Sections 4.1 and 4.2, whilst remaining computationally feasible to explore the effects of the Gaussian representation approach. Instead of updating the voxel intensities directly, this approach represents the unknown 3D object volume as a sum of learnable 3D Gaussian components.

The Simplified Gaussian method kept the same shared preprocessing pipeline and FLFM forward model. The main reason for this was to ensure fairness when evaluating and comparing results. The key design difference for this method was the representation of the unknown 3D object as a sum of learnable Gaussian components instead of a dense voxel volume.

In the simplified Gaussian design, the unknown target object is parameterised using a fixed number of Gaussians, where each Gaussian has a learnable intensity weight, centre location, and a width along the z, y, and x axes. Evaluating these Gaussians over fixed

coordinate grids and summing them estimates the 3D object volume:

$$g_{\theta}(z, y, x) = \sum_{i=1}^M \rho_i \exp\left(-\frac{1}{2} \left[\frac{(z - \mu_{z,i})^2}{\sigma_{z,i}^2} + \frac{(y - \mu_{y,i})^2}{\sigma_{y,i}^2} + \frac{(x - \mu_{x,i})^2}{\sigma_{x,i}^2} \right]\right) \quad (4.4)$$

where M is the number of Gaussian components, ρ_i is the learnable intensity weight of the i -th Gaussian, $(\mu_{z,i}, \mu_{y,i}, \mu_{x,i})$ is its centre location, and $(\sigma_{z,i}, \sigma_{y,i}, \sigma_{x,i})$ are its widths along the z , y , and x axes.

This equation is adapted from Eq. (3) in Xu et al. [3], where each Gaussian is defined by a learnable density, covariance matrix, and mean. In this project, the covariance is assumed to be diagonal, replacing the full covariance matrix with independent widths along the z , y , and x directions and therefore allowing for the removal of the rotational terms. In contrast, in the full 3DGAT method, the covariance allows arbitrary ellipsoidal orientation. After estimating the unknown volume, it is projected through the same FLFM imaging model used in Sections 4.1 and 4.2. The final predicted image can be written as:

$$\hat{O}(u, v) = \sum_{j=1}^Z (H_j * g_{\theta,j})(u, v) \quad (4.5)$$

Which is adapted from Eq. (2) the discrete FLFM imaging model in Xu et al [3].

where H_j is the PSF slice at depth j , $g_{\theta,j}$ is the reconstructed object slice at depth j , $\hat{O}(u, v)$ is the predicted FLFM measurement, and $*$ denotes two-dimensional convolution.

4.4 Design of Simplified 3D Gaussian Adaptive Reconstruction with TV Regularisation Term

The simplified 3D Gaussian adaptive reconstruction with a TV regularisation term was designed as an extension of the algorithm described in Section 4.3. This method extended the simplified Gaussian design in Section 4.3 whilst retaining the shared preprocessing pipeline, Gaussian parametrisation, and FLFM forward model. The main extension of this design was the addition of a TV prior to the optimisation process of the algorithm. Since the simplified Gaussian method is solved through gradient-based optimisation, the TV prior was incorporated as an additive regularisation term in the loss function. The total loss function is defined as

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{ImgDData}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}} \quad (4.6)$$

Where $\mathcal{L}_{\text{ImgDData}}$ is the image-domain data term, $\mathcal{L}_{\text{TV}}$ is the TV penalty applied to the reconstructed volume, and λ_{TV} is the regularisation weight. For this design, the regularisation weight λ_{TV} controls how much the reconstruction prioritises fitting the FLFM or becoming smoother.

4.5 Design of basic XAI application

The basic XAI application was designed as a simple and intuitive interactive interface with the aim of improving the interpretability and usability of the reconstruction algorithms. The purpose of this application was to make it easier to visualise the outputs of the different reconstruction algorithms by uploading the relevant FLFM input image and corresponding PSF file. Once the relevant files were uploaded, a reconstruction method could be selected, producing a reconstruction output in a visually interpretable form. This design approach aligns with the overall project goal of visualising reconstruction outputs through an XAI-based application.

The interface’s design was organised as a pipeline consisting of file input loading, algorithm selection, image reconstruction, Sparrow separability analysis, and finally output display.

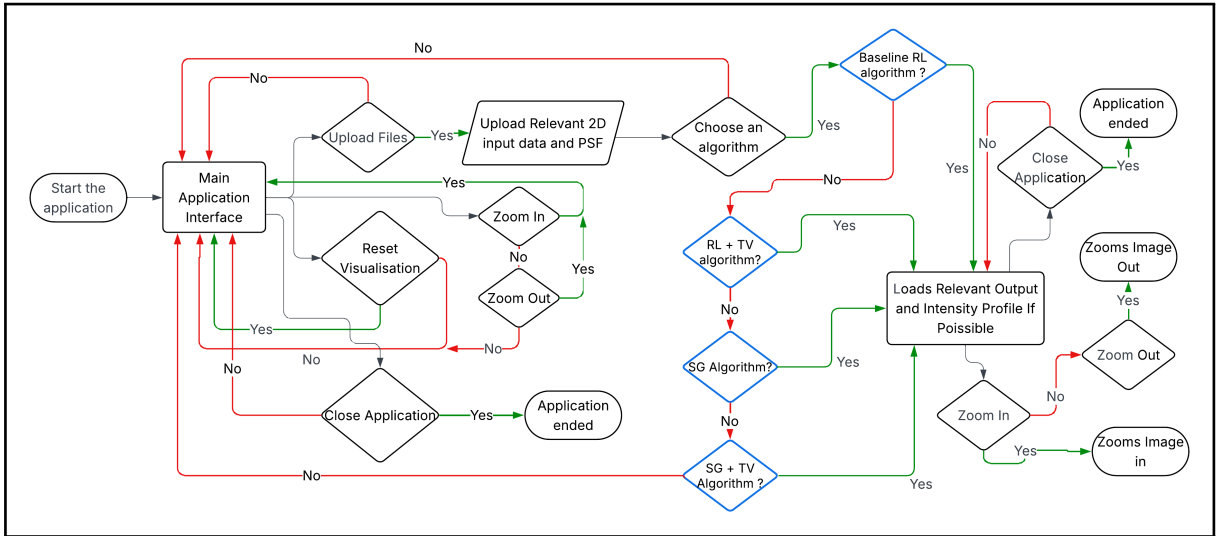


FIGURE 4.1: Flow diagram of the XAI application.

The reason that the pipeline was created this way was so that a logical flow could be implemented instead of presenting only a final image. The interface was developed to show the reconstruction output on the left side and the Sparrow resolvability decision with the associated intensity profile, if resolvable, on the right side. This design choice

helps to allow users to inspect both reconstruction results and supporting evidence of the algorithm’s effectiveness. Zoom in and zoom out functionality also helped to support visual inspection of the image details. Overall, the application represents a basic form of XAI because it enables the algorithm’s behaviour to be more understandable. This is done through a transparent sequence of pipeline stages and clear visual outputs, whilst hiding the full internal optimisation processes of each method.

Chapter 5: Implementation

5.1 Development Environment, libraries and tools

This project was mainly carried out using Python making use of image processing and computing libraries such as NumPy, SciPy and PyTorch. These tools were chosen because they provide additional support for numerical computation, convolution-based image formation, and reconstruction evaluation. These libraries enable faster development and experimentation throughout the project. Additionally, Matplotlib was used for plotting and result visualisations during testing, and scikit-image was used in the evaluation stage for tests such as peak detection and intensity profiling. For the simplified 3D Gaussian reconstruction algorithms, PyTorch was used to define learnable Gaussian parameters, and enable optimisation on available hardware. The local XAI application was implemented using Tkinter, with PIL used for image handling. The user interface layout was first planned in Figma before being translated into Python code. GitHub was used for version control and iterative development, while OneDrive was used as a practical and reliable file storage for large assets from the project that could not otherwise be stored by GitHub.

The overall framework was designed in a modular way such that making changes to one part of the code would not break the rest of the code. One trade-off for choosing Python as the main development programming language was that it is less computationally efficient than other low-level languages such as C or C++, which increased the runtime for iterative reconstruction and high volume data processing. However, its high versatility and large range of supporting libraries outweighed this disadvantage for the goals of this project.

The reconstruction methods were implemented using a shared Python framework to reduce duplication and enable fair comparisons during evaluation. Across all methods, MATLAB input data were loaded using `loadmat`, converted to `float32` for consistency. After this, the inputs were then checked for valid dimensionality and passed through shared preprocessing stages. This included Nyquist-style low-pass filtering to remove high-frequency artefacts and noise, clipping to set negative values to zero, axial reordering when needed, and then PSF normalisation. Reconstructed outputs were then saved in TIFF format for later evaluation. As a result, the main implementation differences between methods were the reconstruction update rules, regularisation terms, and optimisation approaches.

5.2 Implementation of Base Reconstruction Algorithm

The baseline reconstruction algorithm was implemented in Python and organised into a preprocessing helper function, a reconstruction function, and a main execution block. To begin with, MATLAB input data were first imported using `loadmat`, after which the measured image and PSF stack were then converted to `float32` to ensure numerical consistency and reduce unnecessary memory consumption during the iterative processing stage. Slice-wise convolutions were then implemented using `fftconvolve`. This implementation provided an efficient way to evaluate the forward projection and back projection steps of the Richardson-Lucy update because the FLFM imaging model requires repeated 2D convolutions between each depth slice and its corresponding PSF slice. Before reconstruction involving the PSF stacks, `np.moveaxis` was used to ensure the PSF stack order remained consistent. To prevent numerical instability, a small stabilising constant was introduced into the denominator of the ratio-image calculations. Before the main iteration loop, the PSF normalisation term was first computed to prevent unnecessary computations. The main iteration loop then performed 20 update steps, in which a predicted image was formed from the current estimate. This predicted image was then compared against the measured image through a ratio term and back projected through the flipped PSF stack. After reconstruction, the output volume was post-processed and saved as a TIFF stack using `tiffwrite`. This meant that the obtained results could be reused later for quantitative evaluations and visual comparisons.

5.3 Implementation of Base Reconstruction Algorithm with TV Regularisation Term

The baseline reconstruction algorithm with an added TV regularisation term was implemented as an extension of the baseline Richardson-Lucy reconstruction pipeline described in Section 5.2. This method extended the baseline RL implementation in Section 5.2 whilst keeping the same shared input pipeline and FLFM forward model. Compared to the baseline algorithm, the main extension of this implementation was adding a `TV-Regularisation` helper class. This helper class calculated the spatial gradients, divergence, and a curvature-based TV term from the current reconstruction estimate. During each iteration, the curvature-based TV term was evaluated and then combined with a regularisation weight inside the denominator of the multiplicative update. This regularisation weight was set to 0.22 after being selected from a systematic parameter sweep

to provide the strongest balance between reconstruction improvement and numerical stability. Increasing the TV regularisation weight would increase the influence of the TV term and similarly for reducing the weight. This modification of the denominator acted as the implementation of the TV prior, where it would punish excessive variation during reconstruction whilst retaining the same overall forward model as the baseline method. Similar to the baseline implementation, a small constant was added to prevent instability and ensure avoidance of division by zero errors. Furthermore, the reconstruction was clipped to remain non-negative after each iteration, and the final output volume was post-processed through filtering before being saved as a TIFF stack for later evaluation and comparison.

5.4 Implementation of Simplified 3DGAT Gaussian Based Method

The simplified 3D Gaussian reconstruction algorithm was implemented in Python using the PyTorch library as a parameter-optimisation framework. This method reused the same shared preprocessing pipeline. However, the voxel-based update loop was replaced with a parameter optimisation framework using PyTorch. This is where the unknown 3D object was represented by learnable Gaussian components. This method represented the unknown 3D object volume by a `Simplified_3D_Gaussians` class containing learnable Gaussian intensity weights, centre locations, and axis aligned widths in the z, y, and x-directions. This is the main implementation difference compared to previous approaches, where the unknown 3D volume is represented as a dense voxel array. These Gaussian components were evaluated over fixed coordinate grids and summed to form the current volume estimate. A batched FFT convolution-based approach and summations across z-axis slices, similar to Sections 5.2 and 5.3 for the FLFM forward model, were used, in which the estimated volumes would then be passed through. Optimisation was then carried out, with a goal of reducing the loss function using the Adam optimiser, which was done for 300 update steps, with the loss defined as the mean-squared error (MSE) between the predicted FLFM image and the measured image. Before optimisation steps, random seeds were fixed to improve consistent reproducibility and stability. The final reconstructed volume was clipped to ensure non-negativity, then post-filtered using the Nyquist-style low-pass filter, spatially flipped, and saved as a TIFF stack for later evaluation.

5.5 Implementation of Simplified 3D Gaussian Adaptive Reconstruction with TV Regularisation Term

The algorithm was implemented by extending the simplified 3DGAT algorithm described in Section 5.4 by adding a TV regularisation term in the optimisation loop. This method extended the simplified Gaussian implementation in Section 5.4 whilst keeping the preprocessing helper functions, batched forward projection function, Gaussian representation, Adam optimiser, and 300-step optimisation parameters the same.

The key implementation difference compared with the method in Section 5.4 was that, at each optimisation step, a TV loss term was computed from the differences of the current reconstructed volumes in the z, y, and x-directions and added to the MSE data term using a regularisation weight. This meant that instead of updating the Gaussian parameters using the image domain MSE alone, the optimiser updated the Gaussian parameters using a combined loss formed from the MSE data and the TV prior multiplied by its regularisation weight. To ensure a fair comparison with the non-regularised method, only the TV regularisation weight was fine-tuned whilst the rest of the parameters remained the same. The final output volume was then clipped to remain non-negative, post-filtered using the same Nyquist-style low-pass filter, spatially flipped, and saved as a TIFF stack for later evaluation.

5.6 Implementation of basic XAI Application

The basic XAI application’s interface was designed using Figma, extracted into separate images, then imported into Visual Studio Code for implementation. Python was the programming language in which this application was developed, and Tkinter was used as the main GUI library. Using Python meant that the reconstruction algorithms designed previously could be easily integrated into the application’s pipeline. This also meant that it was easier to reuse evaluation logic, improving maintainability and usability throughout the project.

In the implemented interface, controls were added so the user could upload the FLFM input image and corresponding PSF file, select a reconstruction method, and enter the bead separation distance for Sparrow separability analysis. Once the required inputs had been uploaded by the user, the selected reconstruction pipeline was run and the result was then passed to the Sparrow evaluation stage. The interface was then updated to display

the reconstructed image and an intensity profile if resolvable. Reconstructed images were displayed using ImageTk and PIL, and intensity profiles were generated using Matplotlib. Zoom in, zoom out, and reset visuals were also implemented to support closer inspection of the reconstructed images.

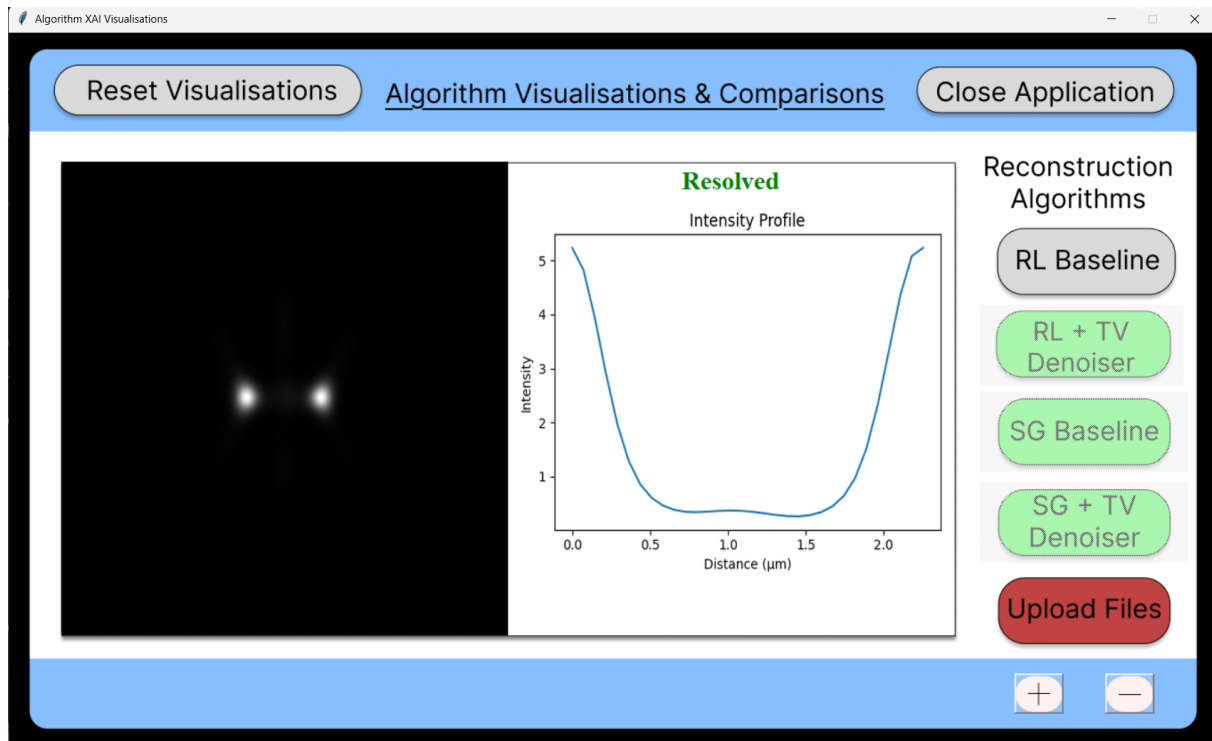


FIGURE 5.1: User interface of the basic XAI application showing the baseline RL algorithm at a reconstructed bead separation of $1.90\ \mu\text{m}$.

Chapter 6: Testing Strategy and Results

6.1 Sparrow Separability Testing

The Sparrow separability test is a way to quantitatively evaluate the effectiveness of the image reconstruction algorithms. The Sparrow criterion defines a minimum distance between two intensity peaks at which the individual features are distinct and separable [17]. This Sparrow resolution criterion can be written as:

$$d_S = 0.47 \frac{\lambda}{\text{NA}}, \quad (6.1)$$

where d_S is the Sparrow resolution, λ is the wavelength, and NA is the numerical aperture. For this project, the Sparrow separability will act as a threshold to determine if two beads are separated or not when the beads are at different distances, when the image is reconstructed. This testing criterion will allow for the comparison of the different algorithms to determine which best identifies when the two beads are identifiable.

For this project, Sparrow separability testing was applied to the reconstructed two-bead data at many different known separations ranging from $0.50 \mu\text{m}$ to $3.20 \mu\text{m}$. After reconstruction, the resulting images were analysed to detect bead peak locations. The pair of peaks whose measured distance was closest to the target separation was selected. This measured distance was then compared against the Sparrow limit criterion to determine whether the two beads remained resolvable. In addition to the quantitative Sparrow threshold, reconstructed images were also checked to validate that the detected peaks corresponded to a physically meaningful bead-pair reconstruction. Detections caused by artefacts outside the expected bead-pair region were not accepted as valid resolvable results.

Beads remaining resolvable indicated that the reconstruction had retained sufficient separation information to detect the two objects as separate, distinct features.

6.2 Full Width Half Maximum Testing

Full width at half maximum (FWHM) is a quantitative measure used to evaluate the axial and lateral localisation performance of image reconstruction algorithms. In this

context, axial localisation is a measure of how spread the reconstruction intensity is when projected in the z-direction, while lateral localisation describes how spread it is in the x and y-directions. Smaller FWHM values mean less spread, which indicates a sharper and better reconstruction, whereas larger FWHM values indicate increased blur or spread. FWHM measures the width of an intensity profile at half of its maximum value and can therefore be used to understand how sharply a reconstructed bead is localised. FWHM is defined as the distance between two points on an intensity profile at which the signal reaches half of its maximum value [18]. This can be written as:

$$\text{FWHM} = x_{\text{right}} - x_{\text{left}} \quad (6.2)$$

where x_{right} and x_{left} are the positions at which the intensity profile reaches half of its maximum value. For this project, FWHM testing was applied to reconstructed single bead data. The bead centre was first identified from the reconstructed volume, after which lateral FWHM measurements were obtained in the x and y directions of the slice containing the peak bead response. The axial FWHM was then measured along the z-direction through the same bead centre. The widths obtained were then converted into micrometres using methods known as lateral pixel spacing and axial slice spacing.

6.3 Model Parameter Selection

Before final benchmarking, key reconstruction parameters were chosen through tuning experiments. For the baseline Richardson-Lucy reconstruction algorithm, representative iteration counts were explored with a goal of finding a stable setting that gave strong reconstruction performance without excessive noise amplification. For the TV-regularised Richardson-Lucy version, the regularisation weight was tuned with a goal to obtain a practical balance between numerical stability and the reconstruction quality. For fair comparison, once the simplified Gaussian reconstruction settings had been chosen through parameter sweeps of the best optimiser settings, iteration count, and Gaussian parameters, these settings were then retained for the TV-regularised Gaussian version. For the TV-regularised Gaussian version, only the TV regularisation weight was adjusted. The final results reported in this chapter therefore correspond to the best-performing parameter settings identified within the tested ranges.

6.4 Sparrow Separability Results

For each reconstruction algorithm, Sparrow separability was first evaluated using a coarse sweep of bead separations, followed by a finer sweep. The initial coarse sweep used wider increments in order to identify the region of transition where the reconstruction results changed from non-resolvable to resolvable. After this region was identified, a finer sweep was carried out using increments of $0.01\ \mu\text{m}$ to estimate the minimum resolvable separation more precisely.

TABLE 6.1: Sparrow separability results across tested bead separations for reconstructions generated at a resolution of 1376×1376 .

Distance (μm)	RL	RL+TV	3DGAT	3DGAT+TV
0.50	N	N	N	N
1.00	N	N	N	N
1.10	N	N	N	N
1.20	N	N	N	N
1.30	N	N	N	N
1.40	N	N	N	N
1.50	N	Y	Y	N
1.60	Y	Y	Y	N
1.70	Y	Y	Y	N
1.80	Y	Y	Y	N
1.90	Y	Y	Y	N
2.00	Y	Y	Y	N
2.50	Y	Y	Y	N
3.00	Y	N	Y	N
3.10	Y	Y	N	N
3.20	Y	Y	N	N

In Table 6.1, Y denotes yes for resolvable and N denotes no for not resolvable.

TABLE 6.2: Summary of Sparrow separability performance for each reconstruction algorithm.

Algorithm	Coarse transition range (μm)	Minimum resolvable separation (μm)
Baseline RL	1.50 to 1.60	1.51
RL + TV	1.40 to 1.50	1.43
Simplified Gaussian	1.40 to 1.50	1.45
Simplified Gaussian + TV	No resolvable transition detected	Not detected within the tested range

Table 6.1 shows that the Richardson-Lucy reconstruction algorithm first became resolvable between $1.50\ \mu\text{m}$ and $1.60\ \mu\text{m}$, while both the TV-regularised Richardson Lucy

method and the simplified Gaussian method first became resolvable between $1.40\ \mu\text{m}$ and $1.50\ \mu\text{m}$. The refined sweep results in Table 6.2 show minimum resolvable separations of $1.51\ \mu\text{m}$ for the baseline RL algorithm, $1.43\ \mu\text{m}$ for the RL + TV algorithm, and $1.45\ \mu\text{m}$ for the simplified Gaussian method. By comparison, the simplified Gaussian method with a TV-regularisation term did not produce a resolvable reconstruction within the tested separation range. Although some detections were produced through automated tests, these were visually identified as artefact-driven responses and therefore were not accepted as valid resolvable results.

At larger tested separations, some anomalous behaviour was also observed. Specifically, the RL + TV method was not identified as resolvable at $3.00\ \mu\text{m}$ but was resolvable again at $3.10\ \mu\text{m}$ and $3.20\ \mu\text{m}$. The simplified Gaussian method exhibited the opposite pattern, being resolvable at $3.00\ \mu\text{m}$, but not resolvable at $3.10\ \mu\text{m}$ and $3.20\ \mu\text{m}$. These results were treated as large-separation inconsistencies rather than conclusions to the minimum resolvable separation results.

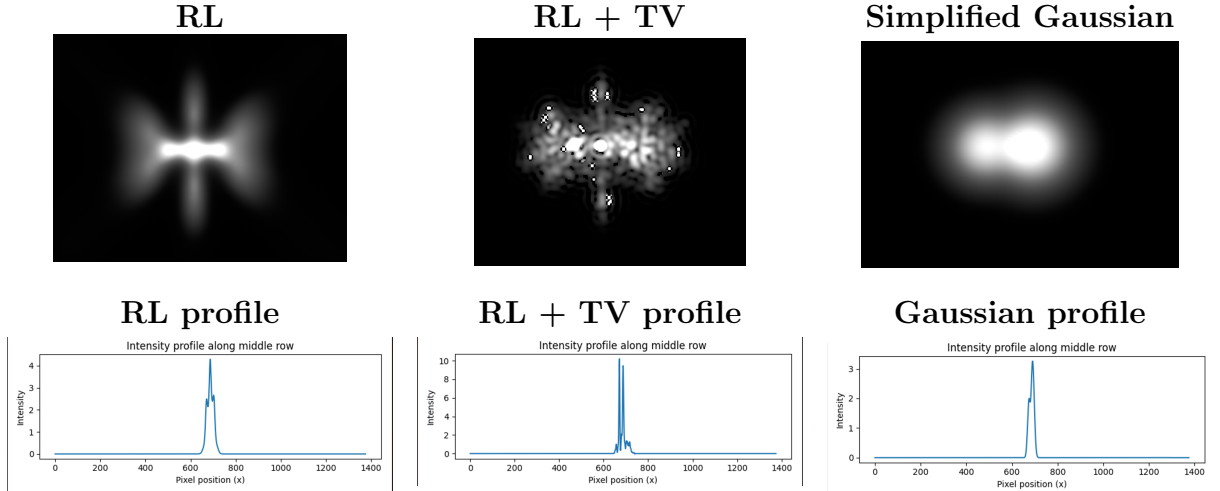


FIGURE 6.1: Reconstructed bead pair images and the corresponding horizontal intensity profiles at the minimum resolvable separation for the three reconstruction methods that produced a separation.

6.5 FWHM Results

From single-bead simulated data, the reconstructions were evaluated to obtain the FWHM measurements. This allowed the localisation sharpness of each method to be analysed. For each reconstructed volume the bead centre was calculated from the maximum intensity projection. Intensity profiles were then extracted through the bead's centre coordinates.

The FWHM was then calculated from these profiles, with smaller values indicating a narrower reconstructed bead response.

TABLE 6.3: FWHM results obtained from single-bead reconstructions at a resolution of 1376×1376 .

Algorithm	$FWHM_x$ (μm)	$FWHM_y$ (μm)	Mean $FWHM_{xy}$ (μm)	$FWHM_z$ (μm)
Baseline RL	0.519	0.519	0.519	2.954
RL + TV	0.305	0.307	0.306	1.305
Simplified Gaussian	0.664	0.664	0.664	3.265
Simplified Gaussian + TV	0.664	0.664	0.664	3.276

Table 6.3 shows that the RL + TV method produced the smallest lateral and axial FWHM values, indicating the sharpest reconstructed bead response among the tested methods. The simplified Gaussian and simplified Gaussian + TV methods produced very similar lateral FWHM values, while the TV-regularised simplified Gaussian method showed a slightly larger axial FWHM. The baseline Richardson-Lucy reconstruction algorithm produced smaller axial and lateral FWHM values than both simplified Gaussian methods, but larger than the values produced by the RL + TV method. These results suggest that TV regularisation improved localisation performance for the Richardson-Lucy reconstruction algorithm, but did not provide a meaningful improvement for the simplified Gaussian method.

6.6 Runtime Results

To compare the computational cost of the four reconstruction algorithms, the runtime performance of each algorithm was evaluated. All timing measurements were obtained using the same hardware described in Section 3.4.1 and at a reconstruction resolution of 1376×1376 . The mean runtime and standard deviation were calculated, with each algorithm repeated five times using the same inputs consistently throughout. This ensured a fair comparison of computational efficiency across methods.

TABLE 6.4: The runtime comparison of the reconstruction algorithms at a resolution of 1376×1376 .

Algorithm	Mean runtime (s)	Standard deviation (s)
Baseline RL	9.876	0.825
RL + TV	10.420	0.134
Simplified Gaussian	111.605	6.440
Simplified Gaussian + TV	138.262	3.180

Table 6.4 shows that the baseline Richardson-Lucy algorithm produced the shortest mean runtime, followed by the RL + TV algorithm. Both simplified Gaussian methods required significantly longer runtimes compared to the Richardson-Lucy reconstruction algorithms. The simplified Gaussian method with TV regularisation produced the highest computational cost overall. Under the same testing conditions, the most computationally expensive methods were the Simplified Gaussian implementations.

Chapter 7: Critical Evaluation of Results

7.1 Evaluation of Sparrow Separability Results

The Sparrow separability results indicate that the RL + TV method provided the strongest performance, achieving the smallest minimum resolvable separation of $1.43\ \mu\text{m}$. Relative to the baseline RL result of $1.51\ \mu\text{m}$, this corresponds to a 5.3% reduction in the minimum resolvable separation. This suggests that the RL framework, paired with a TV regularisation term, improved reconstruction quality enough to improve lateral separability. The simplified Gaussian method also performed strongly, producing a resolvable separation of $1.45\ \mu\text{m}$, which was slightly better than the baseline RL result of $1.51\ \mu\text{m}$, corresponding to an approximate 4.0% reduction of the resolvable distance. Based on these quantitative results, both RL + TV and the simplified Gaussian methods demonstrated improved separability compared with the unregularised baseline RL method. Figure 6.1 shows intensity profiles at the minimum resolvable distance for the different reconstruction methods that produced a detectable Sparrow separation, excluding the simplified Gaussian with TV-regularisation implementation due to failing to produce valid separations. The intensity profiles show distinct peak and trough structures within the expected separation distances. These intensity profiles support the quantitative results and strengthen the interpretation that there has been a successful bead separation.

The anomalous results that were obtained at larger separations for the RL + TV algorithm and simplified Gaussian method may indicate sensitivity in the reconstruction or Sparrow separability detection pipeline. However, they did not change the main conclusion that the RL + TV method demonstrated the strongest minimum resolvable separation.

From the obtained results, a strong finding was that, in the tested simplified Gaussian reconstruction setting, the addition of a TV regularisation term reduced reconstruction quality. The TV-regularised Gaussian method failed to produce a valid resolvable reconstruction within the tested range. Although some detections were produced, these were visually identified as artefact based responses and therefore were not accepted as valid resolvable reconstruction. This means that it performed worse than all other methods. These results suggest that TV regularisation was not beneficial for the simplified Gaussian method under the tested conditions. This is a significant finding because it highlights that regularisation terms do not universally guarantee improved reconstruction quality.

This shows that although it benefited the baseline RL algorithm, it did not transfer effectively to the simplified Gaussian implementation when under the same tested conditions. One possible explanation for the simplified Gaussian method’s reduced reconstruction quality could be that it already implements a smooth parametric prior through its Gaussian representations. Adding the TV prior to an already smooth representation could over-constrain the reconstruction and worsen separability instead of improving it.

However, these results should be considered with respect to the limitations of the experimental setup faced within this project. The chosen reconstruction resolution limited the scope of the experiments, and higher resolutions may change both the performance and threshold values. A finer spatial representation could improve the localisation or separability, specifically for Gaussian-based reconstructions, which may benefit more strongly from higher resolution reconstructions. In addition to this limitation, hardware limits also restricted how broadly the scalability and resolution effects could be explored in this project. To determine whether the Gaussian methods perform more favourably at higher resolutions or with different regularisation strategies, additional work would be required with fewer practical constraints.

7.2 Evaluation of FWHM Results

The FWHM results show that the RL + TV method provided the strongest lateral and axial localisation performance among the tested methods by producing the smallest FWHM values for the respective axes. Compared with the baseline RL result, the RL + TV method reduced the mean lateral FWHM from $0.519\ \mu\text{m}$ to $0.306\ \mu\text{m}$, corresponding to a 41.1% reduction. Additionally, the axial FWHM decreased from $2.954\ \mu\text{m}$ to $1.305\ \mu\text{m}$, corresponding to a 55.8% reduction. These results suggest that for the two RL methods, adding a TV regularisation term significantly improved reconstruction sharpness and depth localisation. By comparison, both simplified Gaussian methods produced wider reconstructed bead responses, indicating overall weaker localisation performance and reconstruction sharpness.

An additional important result was that the TV regularisation did not provide any meaningful improvement for the simplified Gaussian implementation. The simplified Gaussian and simplified Gaussian + TV methods produced the same lateral FWHM values, while the TV-regularised Gaussian method produced a slightly larger axial FWHM. These results suggest that adding a TV regularisation term to the simplified Gaussian method did not sharpen the reconstruction and slightly worsened the depth localisation. In contrast, the RL + TV method showed reductions in lateral and axial FWHM values, indicating

that TV regularisation was more effective when applied to the RL method than when applied to the simplified Gaussian methods.

The FWHM analysis was carried out using reconstructions from single bead simulated data, meaning that it measures localisation sharpness instead of detecting separability between multiple objects. Therefore, it complements Sparrow separability results instead of replacing them. By testing a broader range of performance characteristics, including both lateral sharpness and axial localisation, a stronger evaluation of the algorithm’s effectiveness can be made.

The obtained FWHM results are evaluated in the context of the experimental conditions. Obtaining the FWHM results was computationally demanding due to using a full 3D PSF for the input data. This meant that performing broader and higher resolution tests would require further work without the constraints faced in this project so that explorations into performance trends can be carried out to determine whether they remain consistent under higher resolution reconstruction settings.

7.3 Evaluation of Runtime and Computational cost

The runtime results show a significant difference in the computational cost between the RL and simplified Gaussian approaches. At a reconstruction resolution of 1376×1376 , the baseline RL method achieved the shortest mean runtime of 9.876 s, and the RL + TV method achieved the second shortest mean runtime of 10.420 s. These methods had considerably lower computational costs compared to the simplified Gaussian method, which required 111.605 s, and the simplified Gaussian + TV method required 138.262 s. The extended runtimes for reconstructions indicate that the Gaussian approaches are much more computationally expensive than the RL implementations.

The addition of a TV regularisation prior had only a small effect on runtime for the RL method, increasing the mean runtime by approximately 5.5%. However, the addition of a TV regularisation prior had a stronger effect on runtime for the simplified Gaussian method. A key reason for this difference is the structure of the algorithms. The simplified Gaussian methods use an Adam optimiser with 300 optimisation steps in PyTorch, with repeated forward projections and loss evaluations, whereas the RL methods use 20 multiplicative update iterations with FFT-based convolutions. Due to the Gaussian implementation already having an expensive optimisation process, adding a TV regularisation term increases it further.

These results suggest that the simplified Gaussian methods were less practical for repeated experimentation on available hardware, and instead, the RL methods were more suitable. Due to the Gaussian based implementations having much higher runtimes than the RL methods, performing parameter sweeps and repeated benchmark testing within this project was much more limited. Furthermore, from a computational perspective, the RL methods provided a stronger balance between usability and efficiency, whereas the Gaussian based methods produced higher computational costs.

7.4 Overall Comparison of Reconstruction Algorithms

Based on the results obtained from all three evaluation criteria, the RL + TV method produced the strongest overall performance. This method achieved the smallest minimum resolvable distance of $1.43\text{ }\mu\text{m}$ in the Sparrow separability test amongst all algorithms tested. Additionally, the RL + TV method produced the smallest axial FWHM value in the FWHM tests, indicating the sharpest overall localisation performance. However, visual inspection suggests that the effect of the TV regularisation term was strong, reducing blurring around the reconstructed beads but also potentially introducing over smoothing and, as a result, loss of fine detail alongside shape distortion.

The baseline RL reconstructions still showed visible blurring and distortion around the reconstructed beads, meaning that reconstruction artefacts were not fully removed. The baseline RL method remained the faster approach and therefore retained practical value when prioritising computational efficiency. However, its Sparrow separability and FWHM performance were weaker than the RL + TV algorithm. This suggests that the absence of a TV regularisation term limited reconstruction sharpness and localisation quality. In addition, the simplified Gaussian method showed strong Sparrow separability performance, slightly outperforming the baseline RL method in minimum resolvable separation, but its FWHM and runtime values were worse. Qualitatively, the simplified Gaussian reconstruction appeared to preserve the simulated bead shapes and resembled the input image more closely. This means that, although the simplified Gaussian approach provided an improvement in lateral separability, the trade off came at a higher computational cost without stronger localisation performance.

Ultimately, the weakest performing algorithm was the simplified Gaussian + TV method. It failed to produce any valid resolvable Sparrow reconstruction within the tested range and only noise driven separations. The addition of a TV regularisation term to the simplified Gaussian method did not improve lateral FWHM values and also performed worse axially. This algorithm also produced the highest runtime out of all four algorithms.

These results suggest that TV regularisation improved the baseline RL method but did not work effectively for the simplified Gaussian method.

Overall, the main trade-off identified in this project was between computational cost and reconstruction quality. The baseline RL method was the most computationally efficient, the RL + TV method provided the strongest balance overall, and the simplified Gaussian methods were more computationally expensive alternatives with mixed reconstruction benefits.

Chapter 8: Reflection

8.1 Reflection on the Project Plan vs Actual Progress

The initial plan for this project was to design three reconstruction algorithms. These were a baseline RL algorithm, an RL algorithm extended with TV regularisation, and an RL algorithm extended by adding a CPnP regularisation term. However, the original direction later changed. The revised focus was to understand the effect of the TV regularisation term across two different reconstruction frameworks. These frameworks were a simplified Gaussian method inspired by 3DGAT and a Richardson-Lucy reconstruction method.

These were the main differences from the initial plan. The transition away from a CPnP regularisation term was mainly due to keeping the project feasible with the given time-frame, available computational limits, and implementation complexity. In comparison, the simplified Gaussian approach allowed a broader comparison of reconstruction strategies. At the same time, the changes remained aligned with the overall project aim of evaluating different methods under a shared FLFM reconstruction framework to find an optimal solution that can be used for an AI-enhanced multi-functional metalens microscope.

Another important change to the original scope was the additional testing strategy implemented. Initially, Sparrow separability was the main planned evaluation method, alongside runtime measurements. During this project, one-bead simulated data and FWHM testing were added to provide a stronger understanding of the localisation and reconstruction sharpness. These additions helped to improve the overall evaluation by allowing for a more complete comparison than initially planned.

Some aspects of the project remained consistent with the original direction. The RL and the RL + TV methods were successfully implemented, the XAI visualisation component was developed in the basic form of an app, and Python remained the main programming language throughout, alongside MATLAB for graphical visualisations. The project work stayed shaped by the computational constraints identified during the planning stage.

Overall, the final project aligned with the expected original goals at a high level, but the methods and evaluation strategies changed in response to technical challenges faced and opportunities for stronger analysis. A key lesson from this process was that project plans

must remain flexible. In future work, I would allow for more flexibility in the project plan to account for a potential shift in scope, technical challenges, and testing requirements.

A positive outcome of the project was that the final project structure provided a more meaningful comparison than originally planned, particularly by including simplified Gaussian methods and the additional FWHM evaluation. One difficulty faced in this project was the unexpectedly high runtimes for the Gaussian methods. This made parameter optimisation a long and restrictive process, limiting the ability to explore and optimise parameters as efficiently as the RL methods. Another challenge was the change in algorithm scope. Implementing different algorithm designs that were not initially planned added extra time pressure in the later stages of the project.

Through this process, I strengthened my skills in Python programming, specifically using libraries such as PyTorch, scikit-image, and Tkinter, reconstruction algorithm design, simulation-based testing, user interface design using Figma, and structured technical evaluation. Alongside these technical skills, I also improved my ability to respond to project scope changes with regard to the practical constraints faced. When making these changes, I ensured that they remained aligned with the overall project goals. Finally, I also improved my ability to manage a larger technical project through safer development practices, such as using backups and maintaining version control with GitHub.

Chapter 9: Conclusions and Future Work

9.1 Summary of Contributions, Findings and Limitations

Throughout this project, four different reconstruction algorithms were developed for potential use in an AI-enhanced metalens microscope. The developed methods were a baseline RL reconstruction algorithm, an RL + TV algorithm, a simplified Gaussian algorithm inspired by 3DGAT, and a simplified Gaussian method with TV regularisation. A structured testing pipeline was developed to allow for fair comparisons of the four developed algorithms. This pipeline evaluated the reconstruction quality, localisation performance, and computational practicality. Together, these metrics allowed for a balanced evaluation of the strengths and weaknesses of each algorithm. A further contribution of the project was the inclusion of a basic XAI application framework which was intended to provide visualisation of the reconstruction outputs. Although the application remained a basic implementation, it contributed towards the wider goal of improving the interpretability and accessibility of the reconstruction process. Overall, the project achieved a wider comparison of reconstruction strategies than initially planned and also produced meaningful evidence about the effect of TV regularisation across different reconstruction frameworks.

The results from Section 6 indicated that the RL + TV method provided the strongest overall balance between computational efficiency and reconstruction quality. The RL + TV method achieved the best Sparrow separability result, the strongest axial performance, and a runtime close to the baseline RL method. The simplified Gaussian method demonstrated some competitive Sparrow separability advantages. However, the runtime was significantly longer than the RL and RL + TV methods. The RL method was the fastest in terms of runtime, closely followed by RL + TV. The simplified Gaussian + TV algorithm performed the worst overall. These results suggested that the effect of TV regularisation was dependent on the framework instead of being a guaranteed benefit regardless of the framework. The results obtained from this project showed that it is useful to compare and evaluate reconstruction methods using more than one metric. By combining FWHM values, Sparrow separability, and runtime analysis, it was possible to identify which algorithms performed the best whilst understanding the different trade-offs that each algorithm had. This approach provided a meaningful contribution to better

understand the differences and importance of localisation performance, reconstruction quality, and computational practicality.

Multiple limitations were faced throughout this project. Further algorithm optimisations and testing at higher resolutions were restricted due to hardware limitations. Timing constraints limited evaluation techniques to only Sparrow separability, FWHM, and run-time analysis, meaning other potentially useful techniques, such as the Rayleigh criterion or other artefact analysis methods, could not be explored. These techniques could have provided unique angles for evaluation when judging the different reconstructions' performance. In addition, the evaluation showed that automated separability alone was not sufficient due to the possibility of false positive detections. This meant that qualitative inspection was also required to ensure validity of resolvable detections. The XAI application was also developed in only a basic form, limiting the range of XAI visualisation techniques and interpretability features used for understanding the algorithms on a deeper level. The limitations faced do not diminish the value of the results obtained, but they define the scope for how the project's conclusions should be interpreted.

9.2 Future Work

For future work, the current reconstruction evaluation framework could be extended by including a wider range of testing criteria, specifically methods aimed at understanding qualitative reconstruction accuracy. For example, extra tests to measure artefacts and noise reduction could help to provide a wider understanding of the reconstruction quality. A second consideration for additional work would be to develop the basic XAI application into a more complete interactive tool. This application could be developed further to include visual comparisons between different reconstruction methods, alternative testing measures such as FWHM, alongside visualising the results, and improved support for explaining reconstruction behaviour to non-specialist users.

Ultimately, the different reconstruction algorithms in this project could contribute towards the construction of a fully functional metalens microscope pipeline for cellular imaging. This same general approach could also be applied to a broader range of problems, including non-biomedical applications such as astronomy, where it is important to balance reconstruction quality, computational costs, and interpretability.

Chapter 10: Appendix

10.1 Final Reconstruction Parameter Settings

Table 10.1 summarises the final parameter settings used for the reconstruction results reported in Chapter 6.

TABLE 10.1: Final parameter settings used for the reported reconstruction results.

Algorithm	Iterations / Steps	TV Weight	No. Gaussians	Optimiser	Learning Rate	Nyquist Fraction
Baseline RL	20	N/A	N/A	N/A	N/A	0.20
RL + TV	20	0.22	N/A	N/A	N/A	0.20
Simplified Gaussian	300	N/A	32	Adam	1×10^{-2}	0.20
Simplified Gaussian + TV	300	0.05	32	Adam	1×10^{-2}	0.20

10.2 Evaluation Calculation Settings

Sparrow separability testing used a wavelength of $0.520 \mu\text{m}$, a NA of 0.45, and a pixel size of 100/image width. In Chapter 6, the image width in pixels was 1376. Peaks were detected using `peak_local_max`, from `scikit-image` with thresholds based on the local median and estimated noise level, and the measured peak distance was compared against the Sparrow criterion $d_S = 0.47\lambda/NA$ to determine if a reconstruction can resolve the two beads or not.

FWHM measurements were calculated using an x - y pixel spacing of $100/1375 \mu\text{m}$ per pixel, an axial slice spacing of $0.5 \mu\text{m}$, a lateral half-window of 25 pixels, and an axial half-window of 15 slices.

10.3 Implementation Adaptation

The one-point imaging simulated data used for the single-bead experiments for FWHM evaluations were adapted from a two-point imaging Jupyter notebook provided by Gaochen Dong.

10.4 Algorithm Implementation Listings

This section highlights important workflow logic and unique parts of the different algorithm implementations

```
for i in range(20):
    Predicted_Image = np.zeros_like(0, dtype=np.float32)
    for z in range(Z):
        Predicted_Image += fftconvolve(g[z], H[z], mode="same")
    R = 0 / np.maximum(Predicted_Image, Tiny_C)
    for z in range(Z):
        Depth_Num = fftconvolve(R, H_T[z], mode="same")
        g[z] = g[z] * (Depth_Num / MPD_2D[z])
    np.clip(g, 0.0, None, out=g)
```

LISTING 10.1: Baseline Richardson-Lucy multiplicative update loop.

```
Curvature = TV_Regularisation.TV_Curvature_Term_3D(g, 1e-4)
for z in range(Z):
    Depth_Num = fftconvolve(R, H_T[z], mode="same")
    denom_raw = MPD_2D[z] + Reg_Weight * Curvature[z]
    Denominator = np.maximum(denom_raw, Tiny_C)
    g[z] = g[z] * (Depth_Num / Denominator)
```

LISTING 10.2: TV-regularised Richardson-Lucy update.

```
mini_3DGAT_model = Simplified_3D_Gaussians(
    number_of_gaussians=32, volume_shape=(Z, Y, X)
).to(Device_Usage_Dir)
Optimiser = torch.optim.Adam(mini_3DGAT_model.parameters(), lr=1e-2)

for step in range(300):
    Optimiser.zero_grad(set_to_none=True)
    G = mini_3DGAT_model()
    Predicted_Image = forward_projecting(G, H_torch)
    Loss = F.mse_loss(Predicted_Image, O_torch)
    Loss.backward()
    Optimiser.step()
```

LISTING 10.3: Simplified Gaussian optimisation loop.

```
data_loss = F.mse_loss(Predicted_Image, O_torch)
TV_Z = torch.abs(G[1:, :, :] - G[:-1, :, :]).mean() if G.shape[0] > 1 else G.new_tensor(0.0)
TV_Y = torch.abs(G[:, 1:, :] - G[:, :-1, :]).mean() if G.shape[1] > 1 else G.new_tensor(0.0)
TV_X = torch.abs(G[:, :, 1:] - G[:, :, :-1]).mean() if G.shape[2] > 1 else G.new_tensor(0.0)
Loss = data_loss + (TV_Z + TV_Y + TV_X) * Lambda_TV
```

LISTING 10.4: Simplified Gaussian with TV-regularised loss.

```
def FWHM_1dim_profile(line_profile, peak_height, pixel_size_um):
    line_profile = np.asarray(line_profile, dtype=float)
    if line_profile.size < 3:
        return np.nan
    results = peak_widths(line_profile, [int(peak_height)], rel_height=0.5)
```

```
pixel_width = float(results[0][0])
return pixel_width * float(pixel_size_um)
```

LISTING 10.5: FWHM width calculation helper.

```
def Check_Sparrow(Peak_1, Peak_2, Pixel_Size, Wavelength, NA):
    Peak_Distance = float(np.hypot(Peak_2[0] - Peak_1[0], Peak_2[1] - Peak_1[1]))
    Distance_um = Peak_Distance * Pixel_Size
    Sparrow_um = (0.47 * Wavelength) / NA
    return {
        "distance_um": Distance_um,
        "sparrow_um": Sparrow_um,
        "resolved": bool(Distance_um >= Sparrow_um),
    }
```

LISTING 10.6: Sparrow resolution decision helper.

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